ATTACK DEMONSTRATION & IDS DETECTION RESULTS

University of Cincinnati
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04 | ATTACK OVERVIEW

DEMO CONCEPT

A Kali Linux VM launches real cyberattacks against two identical simulated healthcare networks — one unprotected, one defended by our Suricata + ML IDS. Live alerts demonstrate each attack being classified and flagged in real time while the unprotected network has no awareness of the intrusion.

PORT SCAN

Tool: Nmap

Enumerate open ports and services running on target hosts. Identifies the attack surface and reveals vulnerable application versions on medical device endpoints.

SIGNATURE: High SYN packet rate across multiple destination ports from a single source IP with minimal payload.

DoS / FLOOD

Tool: hping3 / Scapy

Overwhelm target services with crafted traffic to cause a denial of service. Simulates disruption of critical life-support monitoring systems and patient data servers.

SIGNATURE: Massive packet volume spike, repeated identical packet structure, single source, near-zero response rate.

SSH BRUTE FORCE

Tool: Hydra

Automated credential dictionary attack against SSH to gain unauthorized remote access to patient records servers and internal administration accounts.

SIGNATURE: Rapid repeated TCP connections to port 22, sequential auth failure events, single source IP.

ARP SPOOFING / MITM

Tool: arpspoof

Poison ARP cache to intercept traffic between medical devices and the records server. Positions the attacker in-the-middle to capture or alter sensitive data in transit.

SIGNATURE: Duplicate ARP replies from unexpected MAC, MAC-IP conflicts, unsolicited ARP broadcast storm.

05 | ATTACK METHODOLOGY

01 RECONNAISSANCE

The attacker gathers intelligence on the target network using passive OSINT techniques and active probing. IP ranges, hostnames, and running OS versions are identified before launching any exploits.

02 SCANNING

Nmap SYN scans enumerate open TCP/UDP ports across all hosts. Service version detection identifies vulnerable applications such as unpatched SSH daemons or legacy medical device APIs.

03 EXPLOITATION

Identified weaknesses are targeted — brute force attacks against SSH credentials, UDP flood attacks on monitoring services, and ARP poisoning to achieve man-in-the-middle positioning on the local subnet.

04 LATERAL MOVEMENT

After gaining initial access, the attacker pivots — scanning internal segments unreachable from the outside, escalating privileges, and compromising additional hosts on the healthcare LAN.

05 IMPACT & EXFILTRATION

Patient records are exfiltrated over encrypted channels, critical services are disrupted or held for ransom, and tampered sensor data from medical IoT devices is transmitted to clinical staff.

06 | DETECTION RESULTS

ML MODEL PERFORMANCE

Random Forest classifier trained on network flow features — 80/20 train-test split with GridSearchCV hyperparameter optimization and SMOTE oversampling for class balance.

97.2%	96.8%	97.5%	97.1%
Accuracy	Precision	Recall	F1-Score

PER-ATTACK DETECTION RATE

Attack Type	Detection	Alert Rate
Port Scan	99.1%	~120 / min
DoS / Flood	98.4%	~340 / min
SSH Brute Force	96.2%	~45 / min
ARP Spoofing	94.8%	~18 / min

PROTECTED vs. UNPROTECTED

UNPROTECTED

- ✗ No attack detection
- ✗ Silent data exfiltration
- ✗ Services crash unnoticed
- ✗ No incident audit trail
- ✗ Full attacker persistence

PROTECTED

- ✓ Real-time attack alerts
- ✓ Attack type classified
- ✓ Suricata rules triggered
- ✓ Full JSON audit trail
- ✓ Intrusion attempt blocked

KEY TAKEAWAYS

- Our ML-based IDS achieved >94% detection rate across all attack categories with sub-second alert latency.
- The Suricata + scikit-learn pipeline requires no manual signature updates — behavioral anomaly detection adapts to novel threats.
- The VM demo clearly shows the security gap: the protected network generates immediate categorized alerts while the unprotected network is silently compromised.
- Future work: LSTM-based sequence detection for APTs, integration with real hospital network tap, HIPAA compliance alerting.